

30/8/26

Class Assignment - 9

1. FFT (Fast Fourier Transform) - is an efficient algo used to compute the Fourier transform, which converts a signal or image from the spatial time domain $\rightarrow$ frequency domain

low frequency $\rightarrow$ smooth areas

high frequency $\rightarrow$ edges, noise

Key points $\rightarrow$ converts image $\rightarrow$ frequency representation

used in filtering, compression, noise removal

ex - If an image has sinusoidal noise FFT show bright center $\rightarrow$ low frequency
bright spots $\rightarrow$ noise frequency

2. Inverse FFT (IFFT) - IFFT converts frequency domain data back into the spatial domain (original image). after processing (like filtering) we use IFFT to reconstruct the image

Key points - i) Reconstructs original signal

ex - after removing noise in frequency domain - IFFT gives clean image.

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3. FFT shift (centering) - fftshift rearranges the fft output so that the zero-frequency is placed at the center of the spectrum.

shift $\rightarrow$ low frequency $\rightarrow$ center
high frequency $\rightarrow$ edges

It helps in detect noise patterns, easier visualization, easier filtering

4. Inverse fft shift - reverses the effect of fftshift , moving the center back to the original position before applying inverse fft .

we must undo it before reconstruction

key point - required before ifft

ensures correct reconstruction
without it $\rightarrow$ distorted output